

**AMENDMENT NO. 2 JANUARY 2011
TO
IS 14251 : 1995 CAPTAN, TECHNICAL —
SPECIFICATION**

(Page 2, Annex A, clause A-1) — Add the following new clause before A-1:

‘A-0 GENERAL

A-0.1 Either of the two methods, namely, Titration Method (*see A-1*) or GC Method (*see A-3*) shall be used for estimating captan content. However, GC method will be the referee method in case of dispute.’

(Page 2, clause A-2.3) — Add the following new clause after A-2.3:

A-3 GAS CHROMATOGRAPHY METHOD

A-3.1 Principle

The captan, technical is determined by gas chromatography using internal standard technique.

A-3.2 Apparatus

A-3.2.1 Gas chromatograph (GLC) equipped with FID with facilities for on column injection and coupled to a printer-plotter-cum-integrator is used for this determination. The suggested operative parameters are as follows, but can be changed, if necessary, provided standardization is done:

Column	:	Stainless steel, length = 1 m, OD = 1/8 inch, packed with 5 percent SE-30 on Chromosorb W (HP)
Temperature	:	Column oven — 190°C Detector — 260°C Injector — 250°C
Carrier Gas	:	Nitrogen — 30 ml/min Hydrogen — 30 ml/min Air — 315 ml/min
Attenuation	:	16
Range	:	10
Retention time (Guide values)	:	Captan — 4.0 min (Approx) Di- <i>n</i> -butyl phthalate — 3.0 min (Approx)

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A typical GC chromatogram of captan with internal standard is given in Fig. 1.

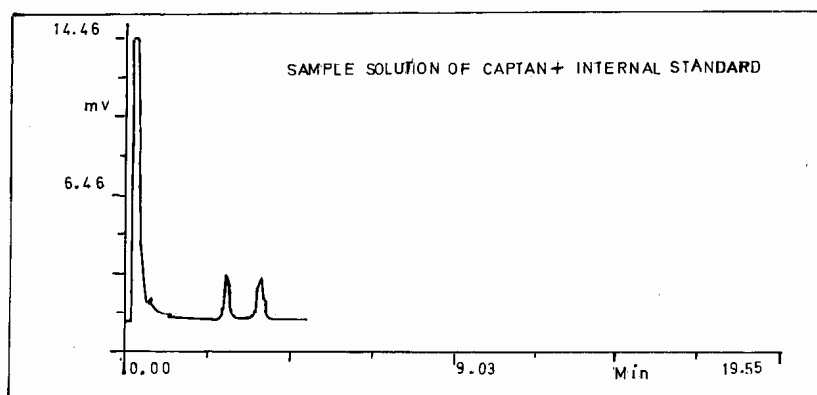


FIG. 1 A TYPICAL CHROMATOGRAM

A-3.2.2 *Injection Volume*, 2 μ l.

A-3.2.3 *Standard Glassware*

A-3.2.4 *Microlitre Syringe*, 5/10 μ l capacity.

A-3.3 Reagents

A-3.3.1 *Acetone*, AR grade or equivalent.

A-3.3.2 *Internal Standard*, Di-*n*-butyl phthalate (DBP) AR grade or equivalent.

A-3.3.3 *Captan Reference Standard*, of known purity.

A-3.4 Procedure

A-3.4.1 *Preparation of Internal Standard Solution*

Weigh out accurately 0.6 g of Di-*n*-butyl phthalate into a 100-ml volumetric flask. Dissolve in acetone and make up the volume up to the mark with acetone.

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A-3.4.2 Preparation of Standard Solution

Weigh out accurately about 0.1 g of captan reference standard of known purity into a 50-ml volumetric flask. Add 20-ml of acetone. Stopper the flask and dissolve using sonicator. Add 10-ml of internal standard solution and make up the volume up to 50-ml mark with acetone. Shake well to homogenize.

A-3.4.3 Preparation of Sample Solution

Weigh out accurately a sample quantity so as to contain 0.1 g of captan into a 50-ml volumetric flask. Add 20-ml of acetone. Stopper the flask and dissolve using sonicator. Add 10-ml of internal standard solution and mix. Add another 20-ml of acetone. Filter and collect the clear filtrate in a stopper test tube.

A-3.5 Estimation

A-3.5.1 Inject Standard Captan Solution (**A-3.4.2**) and Sample Solution (**A-3.4.3**). Measure the areas of captan and internal standard peaks in each case and compute the captan content.

A-3.6 Calculation

$$\text{Captan content, percent by mass} = \frac{R_2}{R_1} \times \frac{M_1}{M_2} \times P$$

where

R_1 = ratio of captan peak area to internal standard peak area of the standard,

R_2 = ratio of captan peak area to internal standard peak area of the sample (whose purity has to be determined),

M_1 = mass in g of captan in standard solution,

M_2 = mass in g of sample taken for test, and

P = percent purity of captan reference standard.

(FAD 1)

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